



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SRI CITY

Brain Computer Interface

Mentor

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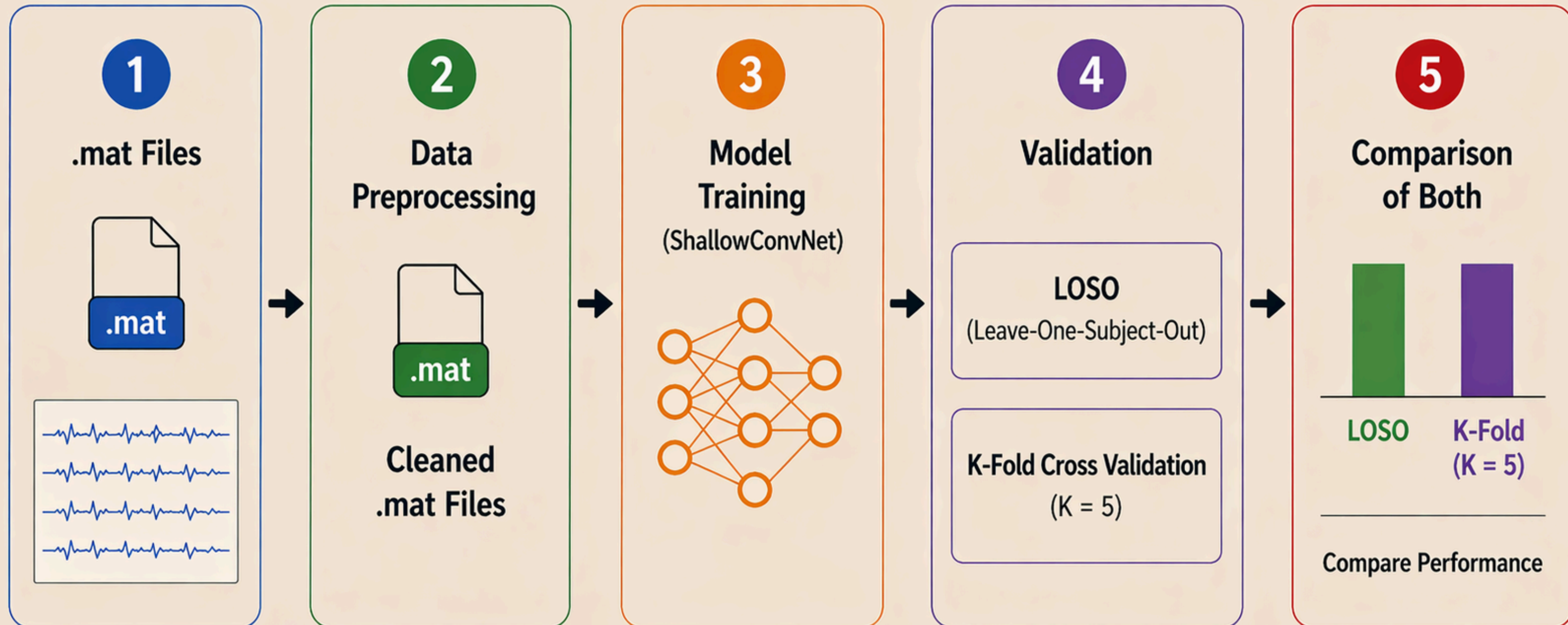
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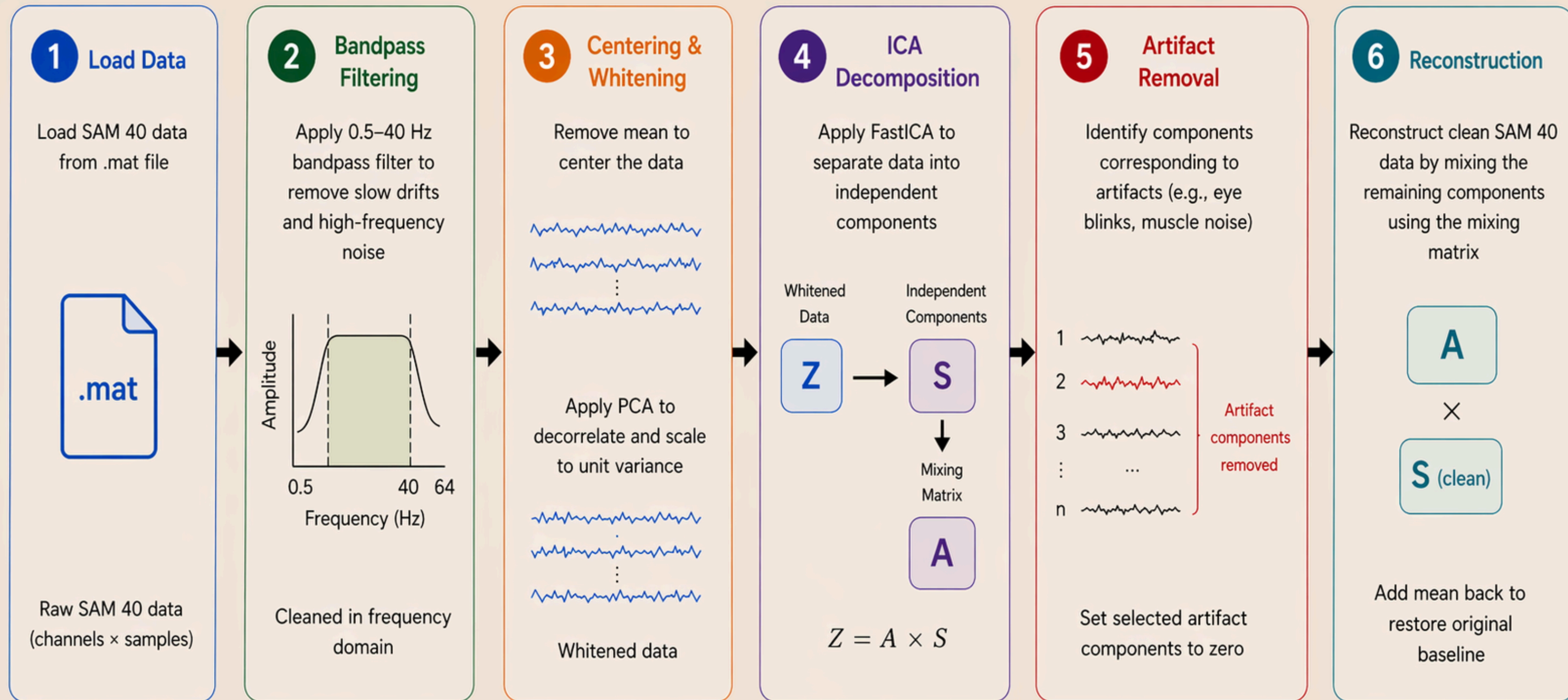
PROBLEM STATEMENT : STRESS DETECTION USING EEG DATASETS

- Stress is primarily extracted mainly from the Prefrontal Cortex (PFC), which is responsible for decision-making, attention, working memory, and emotional regulation.
- We use the SAM40 Dataset for stress detection. [3]
- **DATASET DETAILS:**
 - 40 healthy participants
 - 3 trials per participant
 - Total 120 EEG segments (40×3)
 - 32-channel EEG (10–20 system)
 - Sampling rate: 128 Hz
 - Duration per segment: 25 seconds
 - 3200 samples per channel
 - Dataset shape: (120, 32, 3200)
- **The dataset includes 4 types of induction:**
 - Stroop Color–Word Test
 - Mental Arithmetic
 - Mirror Image Recognition
 - Relaxation State

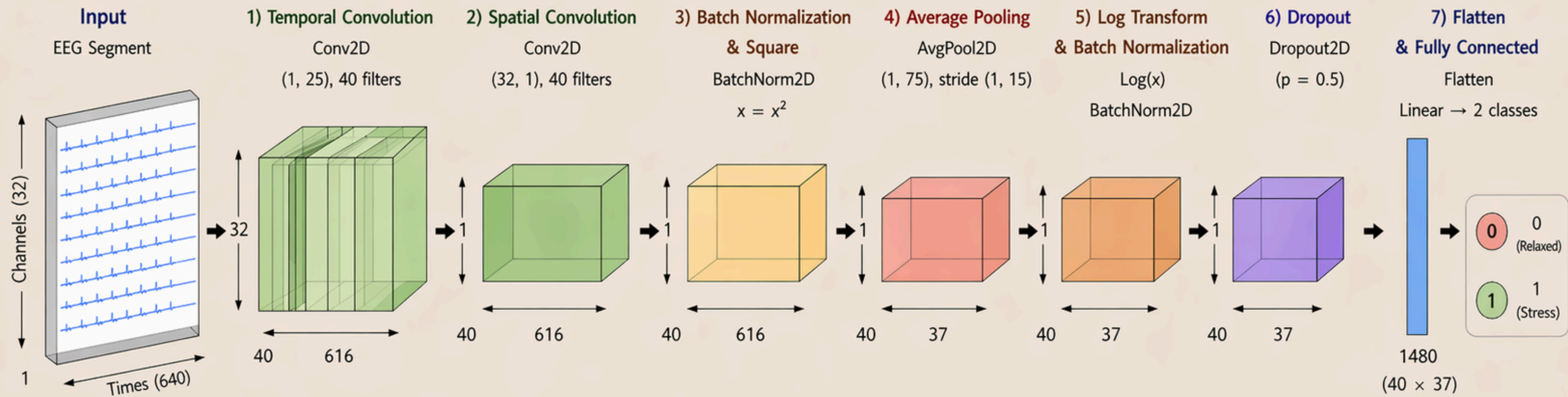
PIPELINE IMPLEMENTED ON SAM40 Dataset



SAM40 DATA PREPROCESSING PIPELINE [1]



ShallowConvNet Architecture [4]



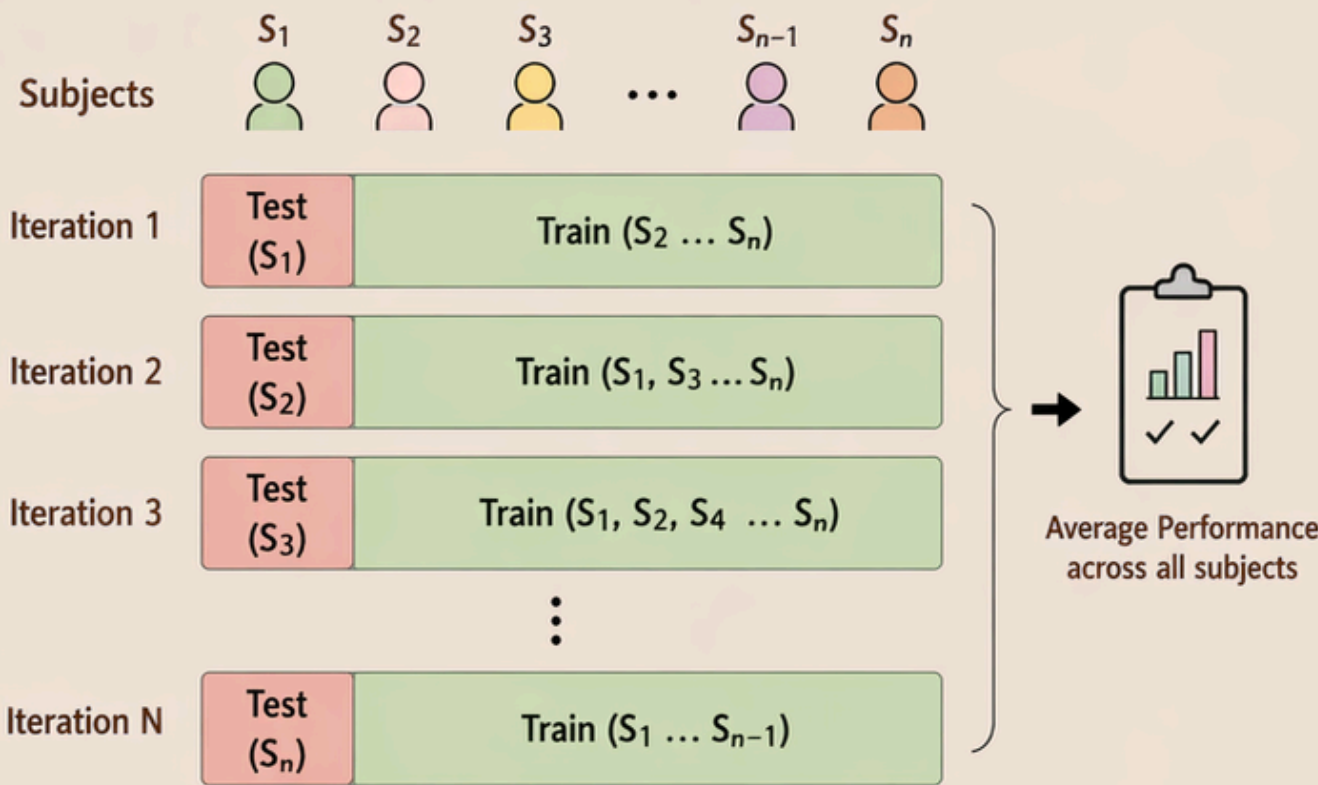
Tensor Shape	(1, 32, 640)	→	(40, 32, 616)	→	(40, 1, 616)	→	(40, 1, 616)	→	(40, 1, 37)	→	(40, 1, 37)	→	(40, 1, 37)	→	(1480)	(2)
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Layer Details	1) Temporal Convolution	2) Spatial Convolution	3) Batch Normalization & Square	4) Average Pooling	5) Log Transform & Batch Normalization	6) Dropout	7) Dense Layer
	- Learns temporal patterns - Kernel spans time - Maintains all channels	- Learns spatial patterns - Kernel spans all channels - Aggregates information across electrodes	- Batch Normalization for faster convergence - Square operation ($x = x^2$)	- Temporal downsampling - Reduces computation - Pool size: 75 - Stride: 15	- Log transform compresses values - Batch Normalization for faster convergence	- Dropout2D - Prevents overfitting - Drop channel maps randomly	- Flatten features - Fully connected layer - Output 2 classes

Cross-Validation Performance Analysis

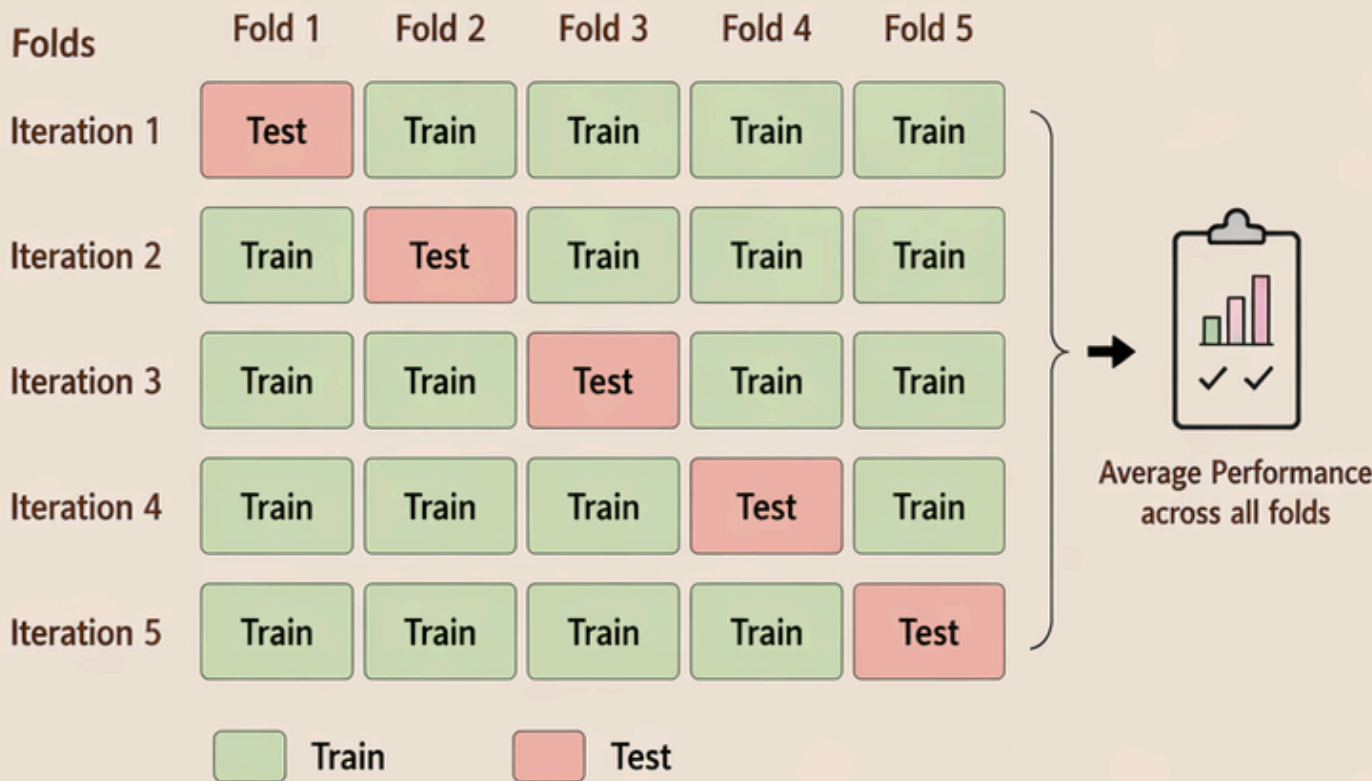
Leave-One-Subject-Out (LOSO)

In each iteration, data from one subject is used as test set and data from all remaining subjects is used for training.



K-Fold Cross Validation (K = 5)

The data is split into K folds. In each iteration, one fold is used as test set and the remaining K-1 folds are used for training.

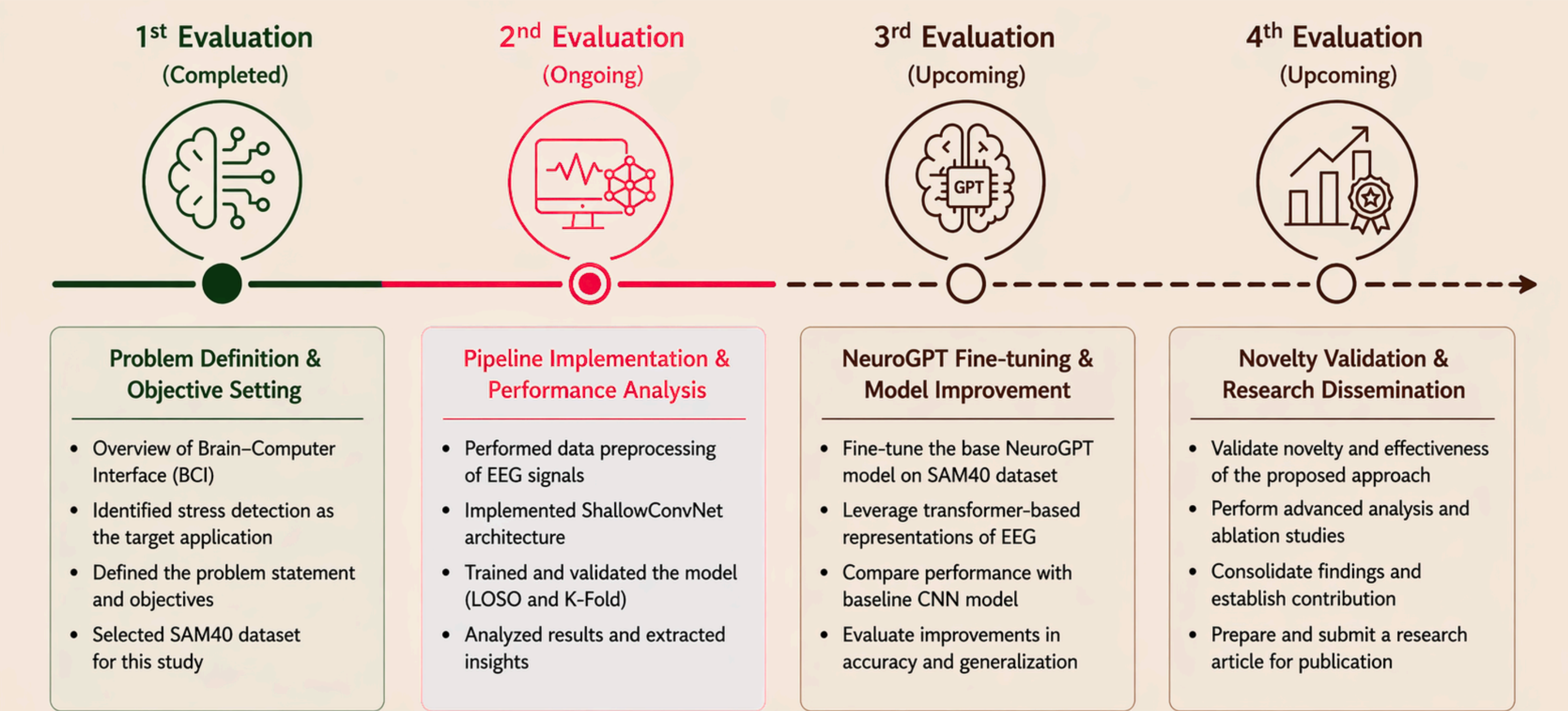


Validation Method	Accuracy (Mean $\pm$ SD)	Balanced Accuracy (Mean $\pm$ SD)
LOSO (Leave-One-Subject-Out)	79.07 $\pm$ 7.94%	72.02 $\pm$ 8.31%
K-Fold Cross Validation (K = 5)	91.46 $\pm$ 1.14%	89.18 $\pm$ 1.26%

Comparison Reports

Article's Accuracy[2]	91.79 ± 2.16%
Our Accuracy	91.46±1.14%

Future Timeline



REFERENCES

- [1]** A. Bablani, D. R. Edla, D. Tripathi, and R. Cheruku, "Survey on Brain-Computer Interface: An Emerging Computational Intelligence Paradigm," *ACM Computing Surveys*, vol. 52, no. 1, Art. no. 20, Jan. 2020, pp. 1–32, doi: [10.1145/3297713](https://doi.org/10.1145/3297713).
- [2]** Abdel-Ghaffar, E.A.; Salama, M. The Effect of Stress on a Personal Identification System Based on Electroencephalographic Signals. *Sensors* 2024, 24, 4167. <https://doi.org/10.3390/s24134167>
- [3]** S. Ghosh, A. Bhowmik, and A. Tibrewal, "SAM 40: Dataset of 40 subject EEG recordings to monitor the induced-stress while performing Stroop color-word test, arithmetic task, and mirror image recognition task," *Data in Brief*, vol. 40, p. 107772, 2022, doi: [10.1016/j.dib.2021.107772](https://doi.org/10.1016/j.dib.2021.107772).
- [4]** R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggenberger, M. Tangermann, F. Hutter, and T. Ball, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017, doi: [10.1002/hbm.23730](https://doi.org/10.1002/hbm.23730).